The Human Heart

Class 11 Biology | NEET | Structure, valves, double circulation, solved numerical and shortcuts

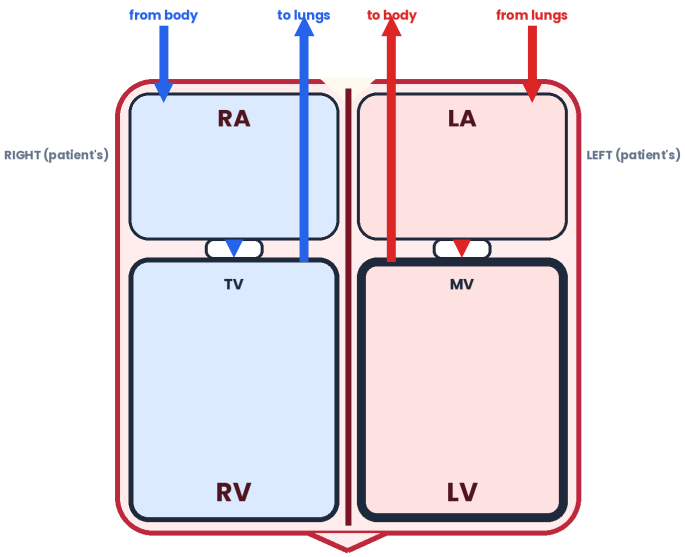
By the end of this chapter you can:

- label the 4 chambers of the heart and state what each does
- explain the valves and why blood flows only one way
- trace pulmonary circulation and systemic circulation (double circulation)
- solve numericals on cardiac output, and use NEET shortcuts for oxygenation status

1. Why the heart matters

Every cell in the body needs oxygen and food, and must get rid of carbon dioxide and waste. Blood carries all of this. The **heart** is a muscular pump that keeps blood moving through the body non-stop. An adult heart beats about **70-75 times per minute**.

2. The four chambers

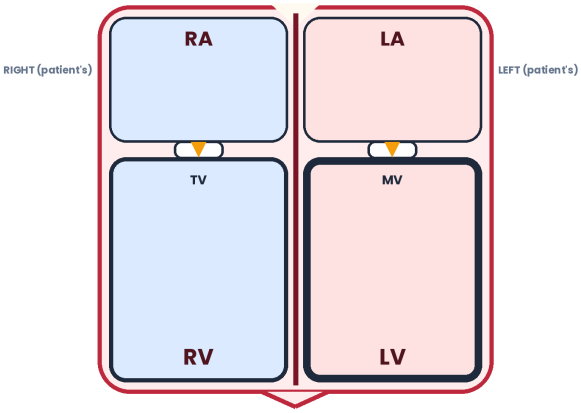


Chamber	Position	Function
Right atrium (RA)	Upper right	Receives deoxygenated blood from the body (via vena cava)
Right ventricle (RV)	Lower right	Pumps deoxygenated blood to the lungs (via pulmonary artery)
Left atrium (LA)	Upper left	Receives oxygenated blood from the lungs (via pulmonary vein)
Left ventricle (LV)	Lower left	Pumps oxygenated blood to the whole body (via aorta) — thickest wall

The two **atria** (upper chambers) receive blood; the two **ventricles** (lower chambers) pump blood out, so ventricle walls are thicker and more muscular. A muscular wall, the **septum**, separates the right side from the left side so oxygen-rich and oxygen-poor blood do not mix.

3. Valves: one-way doors

Valves sit between the atria and ventricles (and at the start of the major vessels). They let blood flow forward, then snap shut to stop it flowing backward. The **“lub-dub”** sound of a heartbeat is these valves closing.



Valve	Location	Cusps
Tricuspid valve	Right atrium → right ventricle	3
Mitral (bicuspid) valve	Left atrium → left ventricle	2

4. Pulmonary circulation (Loop 1): heart <-> lungs

Deoxygenated blood from the body enters the **right atrium** → moves to the **right ventricle** → is pumped through the **pulmonary artery** to the lungs. In the lungs, blood picks up oxygen and releases carbon dioxide. Oxygen-rich blood then returns through the **pulmonary vein** to the **left atrium**.

Exception to remember: the pulmonary artery carries deoxygenated blood, and the pulmonary vein carries oxygenated blood — the opposite of every other artery and vein in the body (arteries usually carry oxygenated blood away from the heart; veins usually carry deoxygenated blood back to it).

5. Systemic circulation (Loop 2): heart <-> whole body

Oxygenated blood from the left atrium moves into the **left ventricle**, which has the thickest wall of all, and pumps it through the **aorta** to the entire body. After delivering oxygen to the tissues, deoxygenated blood returns through the **vena cava** to the right atrium, and the cycle begins again.

Because blood passes through the heart **twice** in one full circuit (once in each loop), this is called **double circulation**.

6. Solved example

Solved example 1

Question	A person's heart beats 72 times per minute, and pumps 70 mL of blood with each beat. Find the blood pumped per minute (cardiac output).
Given	Heart rate = 72 beats/min, Stroke volume = 70 mL
Formula	Cardiac output = Heart rate × Stroke volume
Substitute	= 72 × 70
Answer	**= 5040 mL ≈ 5 litres/min**
Shortcut	72 × 70 = 72 × 7 × 10 = 504 × 10 = 5040 mL

7. NEET shortcut: oxygen status by side

Right side of the heart = deoxygenated blood (like venous blood)

Left side of the heart = oxygenated blood (like arterial blood)

Exception: pulmonary vessels are reversed — pulmonary artery carries deoxygenated blood, pulmonary vein carries oxygenated blood.

Valve memory aid (by cusp count): Tricuspid = right side (3 cusps). Mitral/Bicuspid = left side (2 cusps).

8. Common mistakes

• Mixing up the pulmonary artery/vein exception — remember it is named by which vessel it is (artery/vein), not by what it carries.

• Thinking the left and right atria are connected — the septum keeps oxygenated and deoxygenated blood completely separate.

• Forgetting the left ventricle has the thickest wall, since it pumps blood to the entire body, not just the lungs.

• Saying blood passes through the heart once per circuit — it is double circulation: once per loop, twice per full circuit.

9. Practice questions

Write each answer in board format: Given, Formula, Substitution, Answer with unit.

Q1	A person's heart beats 75 times per minute and pumps 65 mL of blood per beat. Find the cardiac output per minute.
Q2	Name the chamber that receives oxygenated blood from the lungs, and the vessel that carries it there.
Q3	Which valve lies between the left atrium and left ventricle? How many cusps does it have?
Q4	State one way in which the pulmonary artery is an exception to the usual rule for arteries.
Q5	Why does the left ventricle have a thicker wall than the right ventricle?
Q6	Explain, in one line, why the circulation in humans is called 'double circulation'.

10. Answers

Q1	Cardiac output = 75 × 65 = 4875 mL ≈ 4.9 litres/min
Q2	Left atrium (LA); carried by the pulmonary vein
Q3	The mitral (bicuspid) valve; it has 2 cusps
Q4	It carries deoxygenated blood, while most arteries carry oxygenated blood
Q5	It pumps blood through the aorta to the entire body, needing much more force than the right ventricle (which only pumps to the lungs)
Q6	Because blood passes through the heart twice in one full circuit — once via the pulmonary loop, once via the systemic loop

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Notes, solutions and videos from Class 1 to NEET. Watch the video for this chapter and follow on Instagram for daily diagrams and questions.